

Data-Driven Personalised Educational Content Recommendation

CM3070 Final Project · CM3005 Data Science template

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Problem, gap, and five contributions

Problem — ~25% OULAD withdrawal; learners ask for help choosing what to study next.

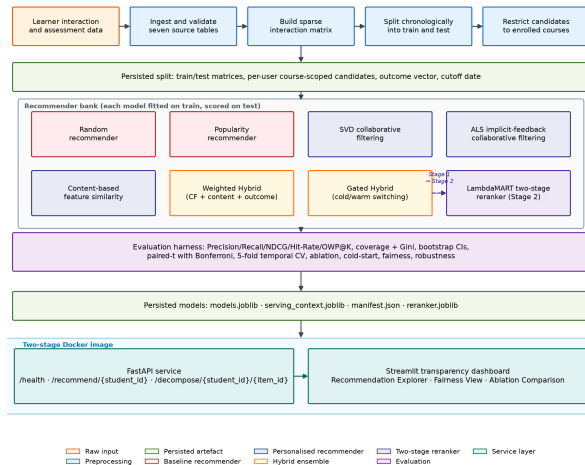
Gap — no public system I am aware of combines CF + content + outcome under an outcome-aware metric.

Five contributions

1. Outcome-weighted Precision@K (novel offline metric)
2. Gated Hybrid — cold/warm switching recommender
3. LambdaMART two-stage reranker with an honest three-window training design
4. FastAPI + Streamlit + Docker serving stack
5. Reproducible statistical harness (CV, bootstrap CIs, paired- t , ablation, robustness)

Pipeline and serving architecture (as delivered)

Figure 3.1 — Pipeline and serving architecture (as delivered)



FastAPI service — three read-only endpoints

- ▶ GET /health — loaded roster, n_students, n_items, cutoff_date
- ▶ GET /recommend/{student_id}?k=10&model=Hybrid
 - ▶ top- K item_ids with rank
 - ▶ Hybrid / GatedHybrid: adds (cf_weighted, content_weighted, outcome_weighted, total)
- ▶ GET /decompose/{student_id}/{item_id} — raw (cf_raw, content_raw, outcome_raw) + weighted breakdown

Course-scoping is enforced server-side; persisted joblib artefacts load once at startup so per-request cost is $O(1)$.

Dashboard · Recommendation Explorer

Educational Content Recommender

API status: ready

API URL: <http://10.0.0.1:8080>

T models loaded

26,014 students

6,358 items

cutoff date: 172

Page

Recommendation Explorer

Gamma Rise

Allison Campaigns

localhost:8501

Recommendation Explorer

Query the recommender for any student in the test set. For the hybrid and fateshybrid models the response includes a per-item score decomposition — the CF (blue), content (orange), outcome (green) weighted contributions summed to the final risk score.

Student ID: 6516

Model: Hybrid

Top 10 for student 6516 (Hybrid)

rank	item_id	cf_weighted	content_weighted	outcome_weighted	total	
1	87599	0.8000	0.8000	0.8000	0.1205	0.8115
2	87591	0.8000	0.8000	0.8103	0.3085	0.2313
3	87592	0.8000	0.8000	0.7614	0.0596	0.7944
4	87592	0.8000	0.8000	0.6345	0.0801	0.6947
5	87594	0.8000	0.8000	0.8000	0.1494	0.3094
6	87592	0.8000	0.8000	0.3412	0.1303	0.2715
7	87592	0.8000	0.8000	0.2473	0.0865	0.3429
8	87576	0.8000	0.8000	0.5396	0.0880	0.9296
9	87575	0.8000	0.8000	0.8913	0.0884	0.3803
10	87587	0.8000	0.8000	0.2996	0.0376	0.3866

Score decomposition

Each bar is one recommendation. height = final hybrid score, colours = CF (blue), content (orange), outcome (green) weighted contributions.

CF Content Outcome

Per-item audit

Explain a specific recommendation

Rank 1 — item 87599

CF (weighted)	Content (weighted)	Outcome (weighted)	Total
0.0000	0.7940	0.1088	0.9028
↑ raw 0.018	↑ raw 0.344	↑ raw/mean score 73.2	

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Dashboard · Fairness View



Dashboard · Ablation Comparison

Educational Content Recommender

API status: ● ready
API URL: <http://localhost:8008>

- 7 models loaded
- 26,074 students
- 6,268 items
- cutoff date: 172

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- ☐ Recommendation Explorer
- ☐ Fairness View
- ☒ Ablation Comparison

Ablation Comparison

Side-by-side top-K from Content, Hybrid, and GatedHybrid for the same student. Where the lists diverge is where the switching behaviour of the Gated variant kicks in.

Student ID

6516

- +

K

Content

rank	item_id	
	1	877059
	2	877032
	3	877042
	4	877076
	5	877085
	6	877069
	7	877087
	8	877062
	9	877092
	10	877066

Hybrid

rank	item_id	total
	1	877059 0.9215
	2	877041 0.2325
	3	877032 0.7946
	4	877042 0.6847
	5	877084 0.1694
	6	877092 0.2725
	7	877062 0.3428
	8	877076 0.5298
	9	877075 0.1803
	10	877087 0.3365

GatedHybrid

rank

Primary temporal split @ K = 10

Model	P@10	NDCG@10	HR@10	OWP@10
Random	0.069	0.069	0.446	0.071
Popularity	0.251	0.272	0.759	0.257
SVD	0.219	0.246	0.799	0.223
ALS	0.218	0.229	0.767	0.222
Content	0.269	0.293	0.790	0.279
Hybrid	0.275	0.303	0.784	0.287
GatedHybrid	0.276	0.303	0.784	0.287

95% percentile bootstrap CIs: P@10 ± 0.004 , NDCG@10 ± 0.004 , OWP@10 ± 0.004 . Paired-*t* Hybrid vs baselines: numerically-zero p-values against Random/Popularity/SVD/ALS on P/NDCG/OWP.

LambdaMART reranker — the honest finding

- ▶ Stage 1: tuned Hybrid retrieves top-100
- ▶ Stage 2: LightGBM LGBMRanker (objective='lambda_rank') reorders via 35 features
- ▶ Honest three-window training: Stage 1 on `tune_train`, labels from `tune_val`, evaluated on `split.test`

Model	P@10	NDCG@10	HR@10	OWP@10
GatedHybrid (deployed baseline)	0.276	0.303	0.784	0.287
LambdaMART (rerank)	0.252	0.284	0.796	0.262

Content + outcome saturates the pointwise-linear space on OULAD's course-scoped catalogue; heavier LTR machinery does not close a gap that is already closed.

Limitations and future work

Limitations

- ▶ OWP is *correlational*, not causal — no interventional study
- ▶ Cold cohort $n = 155$ — wider CI on cold-user precision
- ▶ Gender-fairness gap is descriptive, not diagnosed
- ▶ Single dataset — OULAD only

Future work

- ▶ Fairness-aware re-ranker plugged into /decompose (Ekstrand 2018; Zehlike 2022)
- ▶ Grade-prediction diagnostic against `final_result`
- ▶ Knowledge-tracing signal (BKT/DKT) as a fourth hybrid component
- ▶ Cross-dataset generalisation (MITx / HarvardX)

Thank you

Repo: github.com/VinitTiwari2002/EducationalContentRecommender

Final report PDF · 11 evaluation CSVs · 116 passing tests · Dockerfile